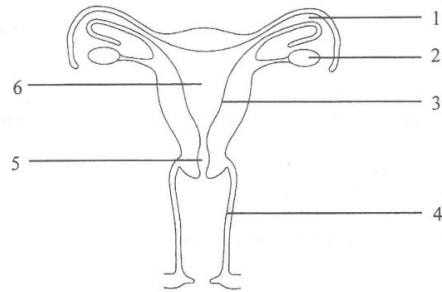


1. 2012Q25

Questions 25 and 26 refer to the following diagram, which shows part of the female reproductive system:



Which of the following structures undergo periodic changes?

- A. 1 and 3
- B. 1 and 4
- C. 2 and 3
- D. 2 and 4

2. 2012Q26

In which of the following structures does implantation normally take place?

- A. 1
- B. 3
- C. 5
- D. 6

3. 2012Q27

Which of the following contraceptive methods prevents the formation of mature gametes?

- A. using diaphragm
- B. using rhythm method
- C. intake of contraceptive pills
- D. using intrauterine device

4. 2012Q28

Which of the following pairs of reproductive structures in humans and flowering plants have the same function?

- | | <i>Humans</i> | <i>Flowering plants</i> |
|----|---------------|-------------------------|
| A. | sperm | anther |
| B. | penis | pollen tube |
| C. | uterus | carpel |
| D. | vagina | petal |

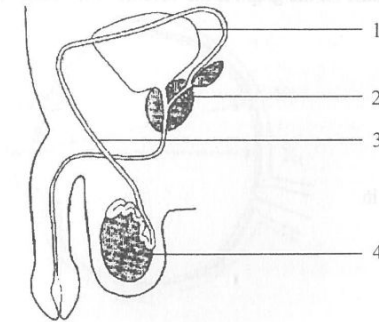
5. 2012Q29

Which of the following is **not** a secondary sexual characteristic of a man?

- A. growth of beard
- B. production of sperms
- C. broadening of shoulders
- D. enlargement of larynx

6. 2013Q35

Questions 35 and 36 refer to the diagram below, which shows the reproductive system of a man:



Which of the following structures are involved in the production of the content of semen?

- A. 1 and 3
- B. 1 and 4
- C. 2 and 3
- D. 2 and 4

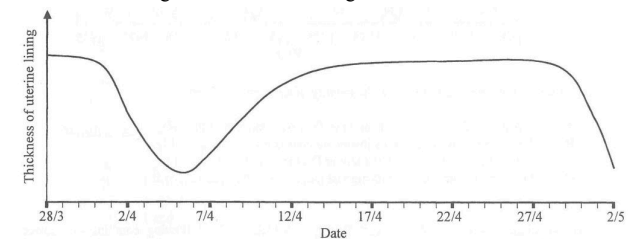
7. 2013Q36

Which of the following may still occur after structure 3 on both sides of the body has been tied up and cut?

- (1) production of sperms
- (2) erection of penis
- (3) ejaculation
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

8. 2014Q28

The diagram below shows the changes in the uterine lining of a woman



During which of the following periods would the woman most likely get pregnant after copulation?

- A. 28/3 to 1/4
- B. 4/4 to 8/4
- C. 11/4 to 15/4
- D. 18/4 to 21/4

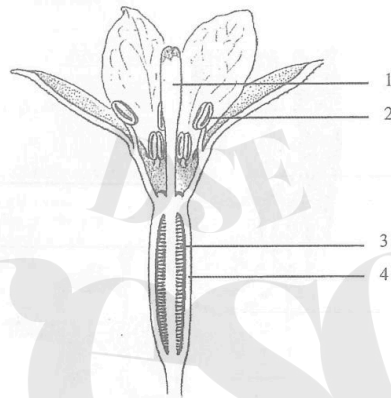
9. 2014Q29

During foetal development, the placenta has functional roles similar to

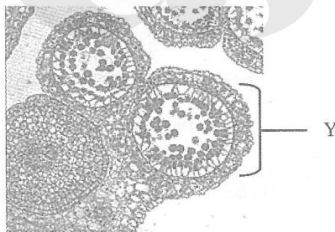
- (1) the bone.
 - (2) the lungs.
 - (3) the small intestine.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

10. 2015Q25

Question 25 refers to the diagram below, which shows the structure of a flower:



The photograph below shows the cross section of structure 2

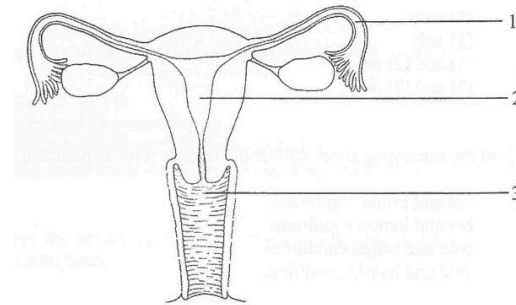


Which of the following parts of the human reproductive system serves a function similar to that of Y?

- A. ovum
- B. sperm
- C. testis
- D. ovary

11. 2017Q19

Questions 19 and 20 refer to the diagram below, which shows some structures of the female reproductive system in humans:



Which of the following combinations correctly shows the location where fertilisation and discharge of semen normally take place?

	Fertilisation	Discharge of semen
A.	1	2
B.	1	3
C.	2	2
D.	2	3

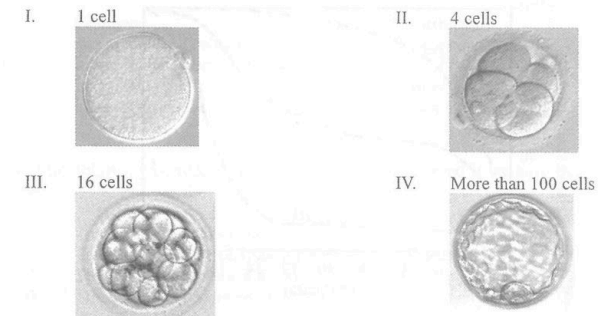
12. 2017Q20

If location 1 is blocked in a 14-year-old girl, which of the following is most likely to occur?

- A. There is no ovulation.
- B. Her fertility decreases.
- C. There is no menstruation.
- D. Her breasts fail to develop.

13. 2017Q23

Question 23 refers to the following photomicrographs of the same magnification. The photomicrographs show some early stages of embryo development:



Which of the following stages is ready for implantation?

- A. I
- B. II
- C. III
- D. IV

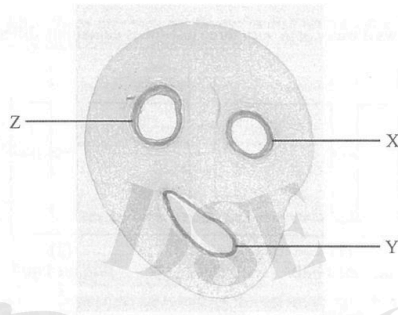
14. 2019Q12

Which of the following statements about the human egg and sperm is correct?

- A. Both have the same number of genes.
- B. Both have the same amount of cytoplasm.
- C. Both have the same amount of food reserve.
- D. Both have the same number of chromosomes.

15. 2019Q27

The photomicrographs below shows a section of a human umbilical cord with three blood vessels:

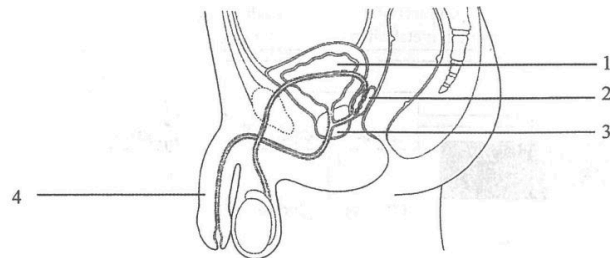


Which of the following comparisons of the content of the blood vessels is correct?

- A. The blood in vessel X has a higher oxygen content than that in vessel Y.
- B. The blood in vessel Z has a higher glucose content than that in vessel X.
- C. The blood in vessel Y has a higher amino acid content than that in vessel X.
- D. The blood in vessel Y has a higher carbon dioxide content than that in vessel Z.

16. 2021Q21

Questions 21 and 22 refer to the diagram below, which shows the male reproductive system and its associated structures:



Which of the labelled structures are responsible for producing seminal fluid?

- A. 1 and 2 only
- B. 1 and 3 only
- C. 2 and 3 only
- D. 1, 2 and 3

17. 2021Q22

With reference to structure 4 above, which of the following structures of flowering plants has a similar function?

- A. style
- B. anther
- C. filament
- D. pollen tube

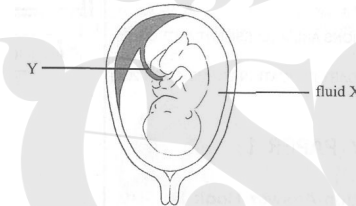
18. 2022Q20

Some babies are born with a heart defect which is a hole in the septum separating the left and right ventricles. When the ventricles contract at the same time, some blood flow will deviate from the normal route. Which of the following is most likely the deviated route?

- A. left ventricle → right ventricle → aorta
- B. left ventricle → right ventricle → pulmonary artery
- C. right ventricle → left ventricle → aorta
- D. right ventricle → left ventricle → pulmonary artery

19. 2023Q34

Questions 34 and 35 refer to the diagram below, which shows a foetus and the associated structures inside the uterus:



Which of the following statements about fluid X is correct?

- A. It is secreted by the uterus.
- B. It provides antibodies to the foetus.
- C. It is a medium for material exchange.
- D. It provides a stable environment for the fetus.

20. 2023Q35

Which of the following combinations correctly compares the blood vessels inside structure Y?

- | | Artery inside Y | Vein inside Y |
|----|-----------------------------------|-------------------------------|
| A. | has a larger lumen | has a smaller lumen |
| B. | carries oxygenated blood | carries deoxygenated blood |
| C. | has less glucose in its blood | has more glucose in its blood |
| D. | blood flows towards foetus' heart | blood flows towards placenta |

21. 2024Q29

Study the following two statements:

- I: Not all copulations lead to fertilisation and pregnancy
- II: Fertilisation occurs when a sperm enters the ovum at the oviduct.

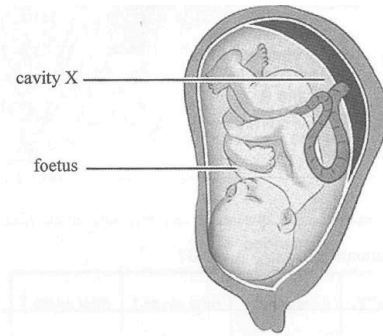
Which of the following descriptions about the two statements is correct?

- A. Both I and II are correct and II correctly explains I.
- B. Both I and II are correct but II does not explain I.
- C. I is correct but II is incorrect.
- D. I is incorrect but II is correct.

LQ

1. 2016Q2

The following diagram shows a foetus and the associated structures inside the uterus:



- a) What is the fluid found in cavity X? (1 mark)
- b) On the diagram, label the structure where the exchange of materials between foetal blood and maternal blood takes place. (1 mark)
- c) Give **two** reasons why foetal blood has to be separated from maternal blood. (2 marks)

2. 2018Q6b

Hong Kong Red Cross Blood Transfusion Service keeps stocks of different blood groups to ensure that there is enough blood supply for transfusion in hospitals.

b) The table below shows some recommendation for adult female and male donors:

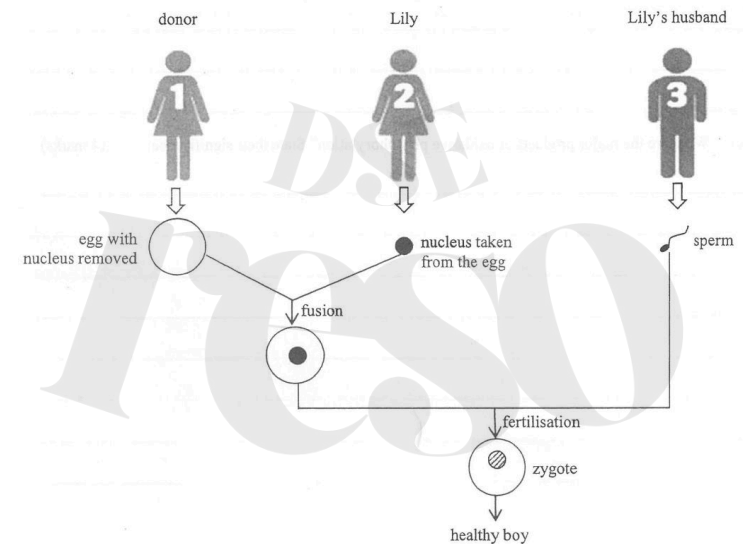
	Female donors	Male donors
Maximum number of donations in a year	4	5
Interval between donations	No less than 105 days	No less than 75 days

Suggest why there are different recommendations for female and male donors. (3 marks)

3. 2018Q9c

Although most DNA is stored in the nucleus, mitochondria also have a small amount of their own DNA. The mitochondrial DNA contains some genes coding for enzymes which are essential for oxidative phosphorylation. Mutations of these mitochondrial genes cause Leigh syndrome, which is a rare and fatal metabolic disease.

- c) Lily had two children who died from Leigh syndrome. It was found that her eggs contained mutated mitochondrial genes. Her children suffered from the disease because normally mitochondria in zygotes come from the eggs while sperms do not contribute any.
- i) With regards to the fertilisation process and the structure of sperms, suggest why sperms do not contribute any mitochondria to zygotes. (1 mark)
- ii) By using a new method called the 'three-parent baby' technique, Lily gave birth to a healthy boy in 2016. Below are the main steps in the technique:



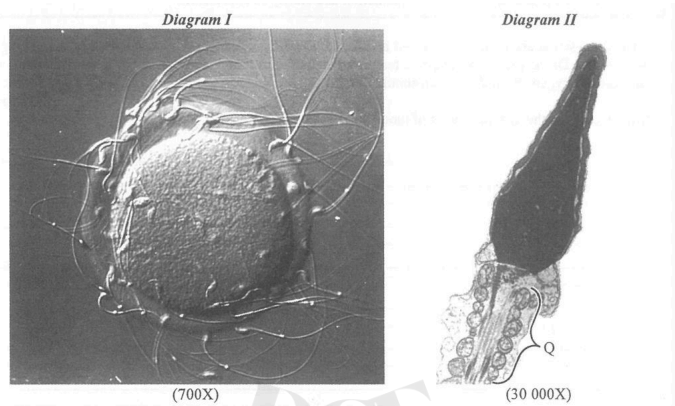
Identify the source(s) of DNA of the nucleus and mitochondria in the boy's cells. (2 marks)

Nucleus:

Mitochondria:

4. 2020Q10

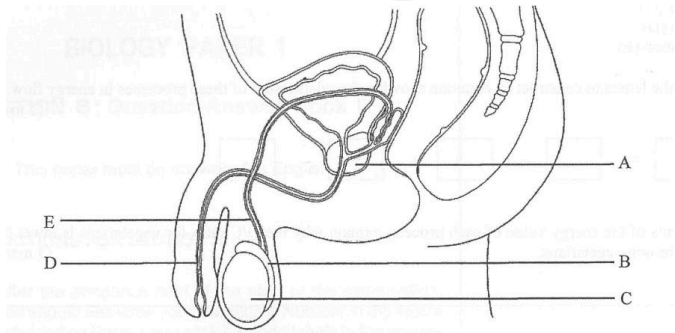
Diagram I shows a photomicrograph of human sperm and a human ovum during fertilisation while Diagram II shows an electron micrograph of an enlarged view of human sperm.



- a) Under normal circumstances, in which structure of the female reproductive system does the process shown in Diagram I take place? (1 mark)
- b) With reference to Diagram II, what is the significance of organelles Q to the sperm's function? (1 mark)
- c) i) Explain the significance of the chromosome number of the sperm and the ovum to sexual reproduction. (2 marks)
- ii) Briefly describe how identical twins may arise after fertilisation. (2 marks)

5. 2022Q2

The diagram below shows the human male reproductive system and its associated structures:



- a) In which structure does meiosis take place? Give your answer using the letters in the diagram. (1 mark)
- b) State the functions of structures A and B respectively. (2 marks)
- c) Vasectomy is an operation for achieving permanent contraception in males.
- i) Using the letters in the diagram, state the structure which is affected in this contraceptive method. (1 mark)
- ii) What is the biological basis of this contraceptive method? (2 mark)

Ans

2012Q25: C (93%)	2012Q26: B (82%)	2012Q27: C (79%)	2012Q28: B (58%)
2012Q29: B (46%)	2013Q35: D (90%)	2013Q36: D (55%)	2014Q28: C (53%)
2014Q29: C (78%)	2015Q25: C (64%)	2017Q19: B (78%)	2017Q20: B (76%)
2017Q23: D (49%)	2019Q12: D (87%)	2019Q27: C (50%)	2021Q21: C (71%)
2021Q22: D (67%)	2022Q20: B (46%)	2023Q34: D (94%)	2023Q35: C (64%)
2024Q29: B (62%)			

2016Q2

- 2 (a) • amniotic fluid (1) (1)
- (b) • placenta with labelling pointing to the correct location (1) (1)
- (c) Any **two** of the following:
- to avoid clumping of blood in case their blood types are incompatible (1)
 - to prevent the entry of some pathogens / toxins from maternal blood directly (1) (2)
 - to avoid breakage of foetal blood vessels by the high blood pressure of maternal blood (1)

4 marks

2018Q6b

- 6 (b) • male donors can donate blood more frequently than female donors or vice versa (1) because
- Any **two** of the following:
- females have regular loss of blood (1) (3)
 - due to menstruation (1)
 - compensatory production of blood cells after blood donation is faster in males

2018Q9c

- 9 (c) (i) • during fertilization, only the head of the sperm, which contains nucleus, will enter the egg / the middle piece of sperm, which contains mitochondria, will not enter the egg / be left outside the egg (1) as a result, the sperm does not contribute any mitochondria to zygote (1)
- (ii) • Nucleus: from Lily and Lily's husband (1) (2)
- Mitochondria: mainly from donor (1)

2020Q10

- 10 (a) in the oviduct (1) (1)
- (b) organelle Q supplies energy for the sperm to swim (1) (1)
- (c) (i) • chromosomes number is haploid (1) in all gametes
- after fertilisation, the chromosomes number is restored to diploid / constant chromosomes number from generation to generation is maintained (1) (2)
- the zygote grows by mitotic cell division into the embryo (1), resulting in cells which are genetically identical (2)
- the embryo is split into two embryos with identical genetic composition at a certain stage (1)

6 marks

2022Q2

- 2 (a) • C (1) (1)
- (b) • A: produces seminal fluid which nourishes the sperms (1)
• B: stores or protects sperm / allows the sperm to mature (1) (2)
- (c) (i) • E (1) (1)
- (ii) • sperm are no longer present in the semen / ejaculated semen does not contain sperm (1)
• there is no fertilisation (1) (2)

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